Computer Vision

HW 2: Feature detection, description, matching

Tyler Mitts

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Introduction

The uses of feature detection, descriptors, and matching are extremely important in many applications of computer vision. From object detection, to motion tracking, to image stitching, the applications of these techniques have a ton of utility. This paper will dive into feature detection, specifically using Harris corners, implementing MOPS feature descriptors from scratch, and different feature matching techniques and the results gained by them. The pipeline used in these experiments goes as follows: image -> autocorrelation matrix -> Harris corners -> MOPS descriptors -> matching techniques -> RANSAC -> match evaluation.

Experiments and Results

Figure 1. Harris corner feature detection of bikes image with minimum distance of 10.

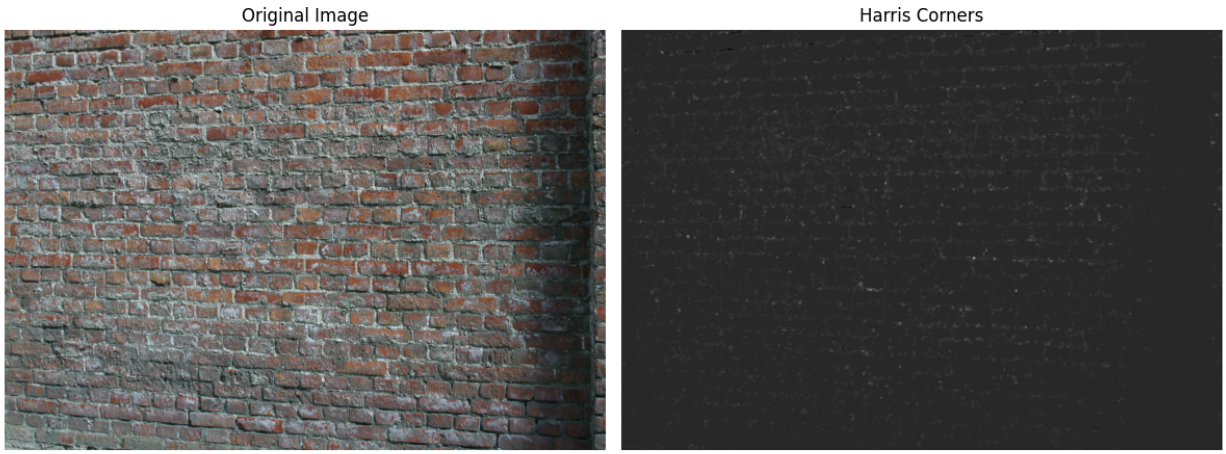

Figure 2. Harris corner feature detection of wall image with minimum distance of 10.

Figure 3. Feature matching from image 1 to image 2 using distance threshold.

Figure 4. Feature matching from image 1 to image 2 using nearest neighbor + threshold.

Figure 5. Feature matching from image 1 to image 2 using distance ratio.

Image 1 to Image 2 (With RANSAC)	Distance Threshold	Nearest Neighbors	Distance Ratio
Threshold used	10.0	10.0	0.7
Number of detected feature points	857	857	857
Number of matched point pairs	112	110	113
True Positive Rate	112/112	110/110	113/113

Figure 6. Comparisons of feature matching from image 1 and image 2, utilizing RANSAC.

Image 1 to Image 2 (Without RANSAC)	Distance Threshold	Nearest Neighbors	Distance Ratio
Threshold used	10.0	10.0	0.7
Number of detected feature points	857	857	857
Number of matched point pairs	246	168	148
True Positive Rate	112/246	110/168	113/148

Figure 7. Comparisons of feature matching from image 1 and image 2, not utilizing RANSAC.

Figure 8. Feature matching from image 1 to image 4 using distance threshold

Figure 9. Feature matching from image 1 to image 4 using nearest neighbor + threshold.

Figure 10. Feature matching from image 1 to image 4 using distance ratio.

Image 1 to Image 4	Distance Threshold	Nearest Neighbors	Distance Ratio
Threshold used	10.0	10.0	0.7
Number of detected feature points	898	898	898
Number of matched point pairs	38	29	6
True Positive Rate	0/38	1/29	5/6

Figure 11. Comparisons of feature matching from image 1 and image 2.

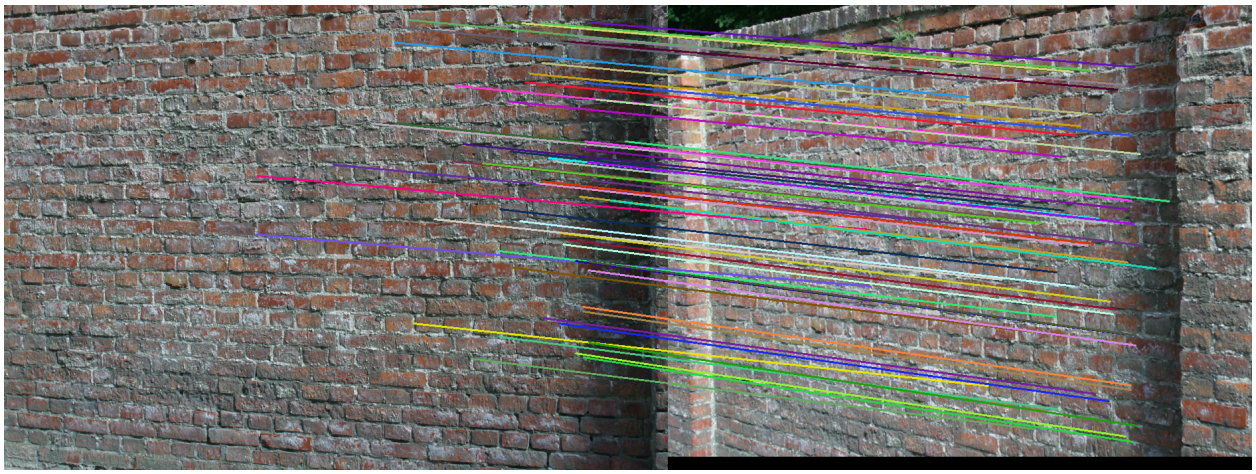

Figure 12. Feature matching from image 1 to image 4 of wall images using distance ratio.

Discussion

When comparing the different matching methods, there isn't too much of a difference when the images are very similar to each other, seen in figure 3-6, all three matching methods worked pretty well post-RANSAC, all having a perfect true positive ratio. The true positives are calculated by comparing the distance of the estimated coordinates, and if it is less than the given threshold (in this case it is set to 3), then it is considered a true positive. Seen in figures 8-11, when the pipeline is tasked to match the image against a blurry, offset version of the image, there is much less success. In both the distance threshold and nearest neighbors methods, when the image is less than ideal, the matching falls apart, with both of them combined only having 1 true positive. On the other hand, the distance ratio still held up, having accurate results through the bad image quality, albeit the number of successful matches significantly decreased, but keeping a TPR of 83%. This shows that the distance ratio is much more robust in comparison to the other methods.

In regards to the TPR, a major reason such good results were found in the first figures was due to the use of RANSAC, a technique used to find the inliers and remove the outliers by selecting random subsets of the data and performing statistics to find points that fit in the most. Seen in figures 6 and 7, RANSAC was able to eliminate all of the outliers, allowing for all of the matching methods to score a TPR of 100%.

For these experiments, the MOPS descriptor was chosen to be implemented so that it would have rotation and illumination invariance, allowing for better matching scores. Seen in the bike images, even if the image is rotated from the original image, it is still able to detect and match the correct points. In figure 12, it displays the MOPS descriptors able to pretty accurately match the points of a wall, in which the camera position has moved to a different angle, and has a

repeated texture, which would make it more difficult for features to be matched correctly. Despite all of that, MOPS in combination with the distance ratio matching method scored a 44/58 TPR.

Overall, the experiments demonstrated the limitations of each of the matching methods, especially showing how the distance threshold and nearest neighbor fall apart as images get further from the image. It also demonstrated the capabilities of MOPS descriptors and being able to quantify the error of the matching based on the real homography between the two images.

Works Cited

Feature Detectors and Descriptors 16-385 Computer Vision Spring 2020, Lecture 6, www.cs.cmu.edu/~16385/lectures/lecture6.pdf.

OpenCV: Feature Matching + Homography to Find Objects, docs.opencv.org/3.4/d1/de0/tutorial_py_feature_homography.html.